Which clinical dataset should Phase 2 use?

Smart Healthcare project | Phase 2 planning | 13 October 2026 | Tanushree Nepal

Summary

- Phase 2 needs ICU clinical data to predict **in-hospital mortality** for cardiac patients from their first 24 hours in the ICU.
- **Full MIMIC-III is not open.** It is free, but only after PhysioNet credentialing (training course, a reference, a signed agreement). Approval takes days to weeks.
- **The Kaggle upload titled "MIMIC-III Clinical Database (Open Access)" is not full MIMIC-III.** We verified its tables are identical to the official 100-patient MIMIC-III *Demo*, minus 22 tables. It is too small and too biased for Phase 2 (Section 3).
- **Recommendation:** build Phase 2 on the open **PhysioNet/CinC Challenge 2012** dataset (12,000 ICU stays, mortality labels, cardiac ICU types) and apply for MIMIC-III credentialing in parallel.

1. What Phase 2 needs

Phase 1 built and evaluated an ECG beat classifier on the MIT-BIH Arrhythmia Database. Phase 2 adds the **clinical** side of the project: vital signs, lab results and patient details from the ICU. Decisions already made for Phase 2:

- **Target:** in-hospital mortality (died in hospital: yes or no).
- **Input window:** the first 24 hours after ICU admission.
- **Cohort:** adult cardiac patients.
- **Variables (from the project brief):** heart rate, blood pressure, SpO_2, respiratory rate, temperature, troponin, CK-MB, creatinine, electrolytes, age, sex.

So the dataset must have ICU admission times, time-stamped vitals and labs, and a mortality outcome, for enough patients to train and test a model honestly.

2. How MIMIC-III access works

MIMIC-III (Beth Israel Deaconess Medical Center, 2001-2012, about 46,500 patients) comes in two official versions on PhysioNet:

Version	Who can download	Size	Usable for Phase 2?
MIMIC-III Clinical Database Demo v1.4	Anyone (open licence)	100 patients, 26 tables, 103 MB	Only for testing code. All 100 patients had died by the time they were selected, so mortality results are meaningless.
MIMIC-III Clinical Database v1.4 (full)	Credentialed users only	about 46,500 patients, about 6-7 GB compressed	Yes, once access is approved.

Steps to get full MIMIC-III (only the student can do these):

- Create a PhysioNet account at physionet.org.
- Complete the CITI course *Data or Specimens Only Research* and download the completion report.
- Apply for credentialing on PhysioNet, naming a reference (usually the supervisor).
- After approval, sign the MIMIC-III data use agreement and download.

Under that agreement the data may not be re-shared or uploaded to outside services, and only aggregate results may be published.

3. Checking the Kaggle "MIMIC-III (Open Access)" upload

Kaggle dataset `ihssanened/mimic-iii-clinical-databaseopen-access` is titled *MIMIC-III Clinical Database (Open Access)*. If it really were full MIMIC-III it would skip credentialing, so we checked what it contains.

Verdict: it is the MIMIC-III Demo, not the full database.

Proof 1 - its own description. The Kaggle page says: *"The MIMIC-III demo provides researchers with an opportunity to review the structure and content of MIMIC-III ... The full dataset is available on PhysioNet."* It lists only 4 tables.

Proof 2 - row counts. Downloaded 13 Oct 2026 (02:15 UTC, 14 Oct) (838,939 bytes; SHA-256 `09191cb1...1b847493`). Only counts and column names were inspected, never individual rows:

File	Rows	What it tells us
PATIENTS.csv	100	Exactly the Demo's 100 patients (full MIMIC-III has about 46,500). **expire_flag = 1 for all 100**: every patient had died, matching PhysioNet's description of the Demo.
ADMISSIONS.csv	129	129 hospital admissions for those 100 patients. 40 died in hospital, 89 survived (31% - typical ICU rates are near 10%).
LABEVENTS.csv	76,074	Lab results for the same 100 patients.
D_LABITEMS.csv	753	Dictionary of lab test names.
structured_medical_records.csv	408	**Not MIMIC data.** Covers 9 patients; the uploader's own simulated medical reports.

Proof 3 - direct comparison with the official Demo. We downloaded the official MIMIC-III Demo v1.4 from PhysioNet and compared each table row by row:

```
official demo tables: 26
ADMISSIONS.CSV: kaggle 129 rows vs official 129 rows | identical columns: True | identical row
content: True
PATIENTS.CSV: kaggle 100 rows vs official 100 rows | identical columns: True | identical row
content: True
LABEVENTS.CSV: kaggle 76074 rows vs official 76074 rows | identical columns: True | identical row
content: True
D_LABITEMS.CSV: kaggle 753 rows vs official 753 rows | identical columns: True | identical row
content: True
ICUSTAYS.CSV: in official demo=True, in kaggle=False
CHARTEVENTS.CSV: in official demo=True, in kaggle=False
DIAGNOSES_ICD.CSV: in official demo=True, in kaggle=False
```

Why it cannot be used for Phase 2:

- **Too small:** 129 admissions; the cardiac subset would be a few dozen.
- **Biased outcome:** every patient was chosen because they had died, so any mortality model or rate would be wrong.
- **Missing tables:** no ICUSTAYS (no ICU start time for the 24-hour window), no CHARTEVENTS (no heart rate, blood pressure or SpO_2), no DIAGNOSES_ICD (no diagnosis codes for a cardiac cohort).
- **One synthetic file** that must not be mistaken for real records.

On the positive side, it is not a leak of the full database, so it raises no licensing problem.

4. The open alternative: PhysioNet/CinC Challenge 2012

An open dataset released for a 2012 competition whose task was exactly ours: predict in-hospital death from early ICU data. It was prepared from MIMIC-II, the predecessor of MIMIC-III.

	Full MIMIC-III	Challenge 2012
What it is	Complete hospital records	Ready-made extract for one task
Access	Credentialed (days to weeks)	**Open - download today**
Patients	About 46,500	12,000 adult ICU stays
Format	26 linked tables to join	One file per ICU stay
Time covered	Whole hospital stay	First 48 hours in ICU
Variables	Thousands	37 time series + age, sex, height, weight
Troponin, creatinine, BP, HR	Yes	Yes
Diagnoses, medications	Yes	No
Mortality label	Derive it yourself	**Provided for all 12,000**
Cardiac patients	Select by ICD-9 code	Select by ICU type: Coronary Care Unit, Cardiac Surgery Recovery Unit
Download size	About 6-7 GB	About 20 MB

Trade-off: Challenge 2012 has no diagnosis codes, medications or raw tables to join, so the brief's table-joining steps (3.2-3.3) are already done for us, and the cardiac cohort is defined by ICU type instead of ICD-9 codes.

5. Other datasets considered

Dataset	Access	What it is	Fit for Phase 2
eICU-CRD Demo v2.0.1	Open	About 2,500 ICU stays, 20 US hospitals	Real schema but still a demo; full version credentialed
PhysioNet Challenge 2019	Open	About 40,000 patients, hourly vitals and labs	Large, but the label is sepsis, not mortality

Dataset	Access	What it is	Fit for Phase 2
MIMIC-IV-ECG	Open	About 800,000 12-lead ECGs, 160,000 patients	ECG side (later phases); clinical linkage needs credentials
PTB-XL	Open	21,799 12-lead ECGs with diagnoses	ECG side only, no clinical data
MIMIC-III / MIMIC-IV / eICU full	Credentialed	Complete ICU databases	Best long-term, after approval

6. Recommendation and next steps

- **Use PhysioNet/CinC Challenge 2012 for Phase 2 now.** It matches the chosen target and is large enough to train and evaluate a model.
- **Apply for MIMIC-III credentialing in parallel.** If approved, add it later as a second data source with the same output format.
- **Do not use the Kaggle upload** or any other unofficial "open" copy of MIMIC.

Sources

- MIMIC-III Clinical Database Demo v1.4 - physionet.org/content/mimiciii-demo/1.4/ (open access, ODbL; 100 patients; 103 MB uncompressed; notes removed).
- MIMIC-III Clinical Database v1.4 - physionet.org/content/mimiciii/1.4/ (credentialed access).
- PhysioNet/CinC Challenge 2012 - physionet.org/content/challenge-2012/1.0.0/ (open access, ODC-By; 12,000 ICU stays; outcomes files Outcomes-a/b/c.txt published for all three sets).
- eICU-CRD Demo v2.0.1 - physionet.org/content/eicu-crd-demo/2.0.1/ (open).
- MIMIC-IV-ECG v1.0 - physionet.org/content/mimic-iv-ecg/1.0/ (open).
- Kaggle - kaggle.com/datasets/ihssanened/mimic-iii-clinical-databaseopen-access.

PhysioNet facts were read from the official pages on 12-13 October 2026. The Kaggle and Demo files were downloaded to a temporary folder, compared, and deleted; no clinical data was added to the project.